

DUAL-MICROPHONE STEERABLE HIGH-ORDER NEURAL DIFFERENTIAL BEAMFORMER

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ABSTRACT

Linear arrays of omnidirectional microphones produce beampatterns that are symmetric about the array axis. For these arrays, a beamformer is considered steerable if its beampattern maintains the same shape in the semicircular plane across all look directions from 0° to 180° . For dual-microphone arrays, conventional differential beamformers are generally non-steerable and restricted to first-order, which significantly limits spatial selectivity. To address these limitations, this study presents a neural differential beamformer (NDBF) with a dual-microphone array. The contributions are as follows: (i) NDBF is steerable; (ii) NDBF achieves high-order frequency-invariant beampatterns; and (iii) NDBF enables stereo recording using only two closely spaced omnidirectional microphones. Experimental results demonstrate that NDBF outperforms existing methods while overcoming the limitations of classical differential beamforming.

Index Terms— Neural Beamformer, Dual-microphone Array, Steerability, High-order Beampattern.

1. INTRODUCTION

Recently, differential beamformers for linear arrays have received significant attention in differential microphone array (DMA) research [1–3]. The largest order of such beamformers is $Q - 1$, where Q denotes the number of microphones [1]. A dual-microphone array, which constitutes the minimum number of microphones for a linear array, restricts the DMA to first order and substantially limits spatial filtering capability.

The steerability of beamformers for a circular array is readily attainable and has been extensively studied [4–6]. In contrast, most differential beamformers designed for a linear array are generally assumed to have a look direction at 0° (endfire) [1, 2], as this configuration typically yields the highest directivity factor (DF). However, the ability to steer the look direction away from 0° while maintaining the same beampattern is essential for practical applications of a linear array, including televisions, tablets, and laptops. For a linear array, “steerability” refers to the ability of a beamformer to maintain a consistent beampattern across look directions from 0° to 180° in the semicircular plane, as described in [3].

Two situations fall short of full steerability: (1) *partially steerable*, where the beampattern achieves 0 dB at the look direction and does not exceed 0 dB elsewhere, but varies across look directions; and (2) *non-steerable*, where the beampattern achieves 0 dB

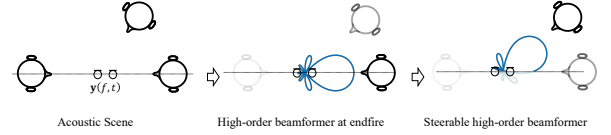


Fig. 1. The goal of the steerable high-order neural differential beamforming with dual-omnidirectional microphones.

at the look direction, but exceeds 0 dB in some other directions [3]. For a differential beamformer with a linear array, achieving steerable beampatterns is feasible only with high-order designs and requires that the null positions satisfy specific conditions [3]. Therefore, a first-order differential beamformer with dual microphones is hardly steerable. Recent linear superarrays [7–10] enable differential beamformers to achieve steerability, but their core concept relies on a hybrid usage of omnidirectional and directional microphones. Therefore, developing a steerable differential beamformer for a linear array, particularly with only two omnidirectional microphones, remains an open research problem.

Most deep neural network (DNN)-based methods have been proposed for performing spatial filtering within an angular region [11, 12] or directly focusing on speech separation or extraction [13, 14], where the beampattern is not explicitly controlled. Recently, neural directional filtering (NDF) [15] proposed to estimate a single-channel mask for nonlinear spatial filtering based on a predefined beampattern using a circular array. This approach paves the way for beampattern-oriented neural spatial filtering and has the potential to address the limitations in differential beamforming. However, NDF has so far been studied only for circular arrays, and its single-channel masking strategy may not fully exploit the spatial information available for beamforming, particularly for dual-microphone arrays, where spatial degrees of freedom are very limited. As an extension of NDF to beamforming for a dual-microphone array, we propose a neural differential beamformer (NDBF) that, using a DNN, produces beampatterns similar to those of classical DMAs. Experimental results demonstrate that NDBF with a dual-microphone array achieves steerable high-order beampatterns and surpasses existing methods. Leveraging the steerability and beampattern-oriented spatial filtering of NDBF, we demonstrate that stereo recording can be achieved with only two closely spaced omnidirectional microphones.

2. PROBLEM FORMULATION

We consider a dual-microphone array comprising two omnidirectional microphones to capture N sound sources in an anechoic environment. The corresponding microphone array signals in short-time Fourier transform (STFT) domain are denoted by $\mathbf{y}(f, t) = [Y_1(f, t), Y_2(f, t)]$, where f and t denote the frequency and time

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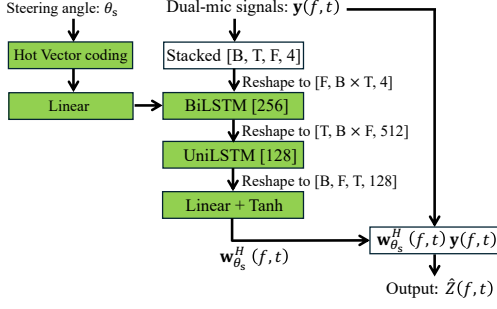


Fig. 2. Extended version of the JNF-SSF architecture [16], which outputs two complex weights per time t and frequency f .

indices, respectively. The mixture signal on the first microphone $Y_1(f, t)$ can be decomposed as

$$Y_1(f, t) = \sum_{n=1}^N X_{1,n}(f, t) + V_1(f, t), \quad (1)$$

where $V_1(f, t)$ is spatially uncorrelated sensor noise and $X_{1,n}(f, t) = H_{\mathbf{p}_1,n}(f) X_n(f, t)$, where $H_{\mathbf{p}_1,n}(f)$ models the direct-path transfer function (DPTF) between the n -th source $X_n(f, t)$ and the first microphone at position $\mathbf{p}_1$.

For this dual-microphone array, the goal of the NDBF is to capture the N sound sources using a steerable high-order DMA beampattern (see Fig. 1). This beampattern is denoted by $\Lambda_{\theta_s}(\theta_n)$, where θ_s is the steering direction of the beampattern and θ_n denotes the direction-of-arrival (DOA) of the n -th source with respect to the position $\mathbf{p}_c$ (the center of the dual-microphone array). The target signal of the NDBF is expressed as

$$Z_{\theta_s}(f, t) = \sum_{n=1}^N \Lambda_{\theta_s}(\theta_n) H_{\mathbf{p}_c,n}(f) X_n(f, t), \quad (2)$$

where $H_{\mathbf{p}_c,n}(f)$ models the DPTF between the n -th source $X_n(f, t)$ and the position $\mathbf{p}_c$. In this work, we propose a DNN based method to obtain the beamformer weights $\mathbf{w}_{\theta_s}(f)$ given any steering direction θ_s at inference.

3. PROPOSED METHOD

3.1. DNN Architecture and Loss Function

The JNF-SSF [16] is adopted as DNN architecture for the NDBF task, with a modification in which the single-channel complex mask is replaced by a vector of complex weights at the output of the final linear layer (see the left side in Fig. 2). The real and imaginary components of the dual-microphone signals in the STFT domain are stacked along the channel dimension, resulting in an input with dimensions of $[B, T, F, 4]$, where B indicates the batch size, T represents the number of time frames, and F denotes the number of frequency bins. The stacked inputs are processed by two distinct long short-term memory (LSTM) modules. The first LSTM is bidirectional and operates along the frequency dimension, thereby modeling instantaneous spectro-spatial relationships in the input [17]. Its output is subsequently processed by a unidirectional LSTM module that captures temporal relationships, treating the frequency dimension as the batch dimension and thus modeling all frequencies independently [17]. As in [16], the desired look direction of the beampattern is encoded as a one-hot vector, which is mapped to an

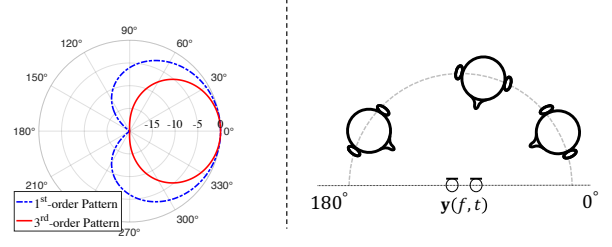


Fig. 3. Target beampatterns (left side) and a source-array setup example (right side).

embedding of the same dimension as the first LSTM hidden states via a linear layer. The output of this layer is then used to initialize the LSTM's states for each time frame. Finally, a linear layer with a hyperbolic tangent activation function computes the NDBF weights $\mathbf{w}_{\theta_s}(f)$. These weights are applied to the microphone array signals $\mathbf{y}(f, t)$ to obtain an estimate of the target signal, i.e., $\hat{Z}_{\theta_s}(f, t) = \mathbf{w}_{\theta_s}^H(f) \mathbf{y}(f, t)$.

The training loss is based on a batch-aggregated normalized $\mathcal{L}_1$ -loss function [6, 18, 19], expressed as:

$$\mathcal{L}_1 = \frac{\sum_{b=1}^B \|\mathbf{z}^b - \hat{\mathbf{z}}^b\|_1}{\sum_{b=1}^B \|\mathbf{z}^b\|_1 + \epsilon}, \quad (3)$$

where the time-domain signals $\hat{\mathbf{z}}$ and $\mathbf{z}$ correspond to STFT representations $\hat{Z}$ and Z , respectively.

3.2. Training Strategy

A J th-order DMA beampattern [20] is defined as follows $\Lambda_{\theta_s} = \sum_{j=0}^J a_j \cos^j(\theta - \theta_s)$, where θ_s represents the steering direction of the pattern, while a_j , where $j \in \{0, 1, \dots, J\}$, are real-valued coefficients that define the pattern's shape, particularly affecting the sidelobes. In sound capture scenarios, primary attention is given to controlling the mainlobe of the patterns. Therefore, a simplified DMA pattern is considered as follows:

$$\Lambda(\theta) = (\mu + (1 - \mu) \cos(\theta - \theta_s))^J, \quad (4)$$

where μ is a real-valued parameter within $[0, 1]$ that determines the null position. The patterns described by (4) emphasize mainlobe shape, which is governed by the order J and μ . In this paper, two example beampatterns were employed as the target beampatterns in Fig. 3. The first target pattern is a 1st-order Cardioid with $\mu = 0.5, J = 1$. The second is a 3rd-order Cardioid pattern with $\mu = 0.5, J = 3$.

To learn the DMA beampattern with a linear array in the free field, the sound sources are positioned along a semicircle, with the microphone array located along the 0° - 180° axis and being concentric with the semicircle (an example is shown in Fig. 3). To simulate a source-array setup with N source positions, N source positions are randomly selected on the semicircle, and DPTFs are simulated for all microphones and sources using the RIR generator [21] with a reflection order of zero. Microphone array signals are then obtained using (1). For each source-array setup, we simulate M target signals for look directions uniformly spanning 0 to 180 degrees, resulting in a steering angular resolution of $\frac{180^\circ}{M}$. The m -th target signal $Z_{\theta_s^m}[f, t]$ corresponding to the steering direction θ_s^m is obtained using (2). During training, we treat microphone-array signals from each acoustic scene, paired with a target signal, as one training sample.

Table 1. SDR (dB) performance: comparison of proposed method with baseline for the end-fire direction.

Method	1 st -order Pattern	3 rd -order Pattern
DMA [1]	-0.99	N/A
Parametric spatial filter [26]	13.77	10.32
NDF [15]	25.85	23.13
Proposed NDBF	25.93	23.24

4. EXPERIMENTAL SETUP

4.1. Datasets

Speech signals from the ‘train-clean-360’ and ‘dev-clean’ subsets of the LibriSpeech corpus [22] served as source signals for training and validation, respectively. For the test sets, speech signals were selected from the EARS dataset [23] using a minimum loudness of -42 dBFS [24]. The parameter M was set to 36, resulting in a steering resolution of 5° . A total of 1440 random source-array setups were simulated, yielding 1440×36 samples for the training set. The test set contains 3240 samples. Each sample in all datasets has a duration of 4 seconds. In each sample, the concurrent sources were randomly drawn from the setup’s source positions, with up to three for training and exactly two for testing. Source positions for training and validation were selected from $\theta_n \in \{0^\circ, 5^\circ, \dots, 175^\circ\}$ and $\theta_n \in \{2.5^\circ, 7.5^\circ, \dots, 177.5^\circ\}$, respectively. The source positions in test sets were selected from $\theta_n \in \{1.25^\circ, 3.75^\circ, \dots, 178.75^\circ\}$. The distance between the two microphones was set to 3 cm, and the source-array distance for all speakers was fixed at 1.5 m. Microphone sensor noise was added with an SNR of 30 dB. STFT settings and training details followed NDF [15].

4.2. Performance Measures

Estimated Beampattern: For each test sample, the array signals for the n -th source is denoted by $\mathbf{x}_n(f, t) = [X_{1,n}(f, t), X_{2,n}(f, t)]$. Then, the estimated beamformer $\mathbf{h}(f)$ is applied separately to the array signals for each source. Subsequently, the wideband power ratio $\xi[\theta_n]$ for the n -th source is calculated as follows:

$$\xi[\theta_n] = \frac{\sum_{f=1}^F \sum_{t=1}^T |\mathbf{h}^H(f) \mathbf{x}_n(f, t)|^2}{\sum_{f=1}^F \sum_{t=1}^T |X_{1,n}[f, t]|^2}. \quad (5)$$

The arithmetic mean of the ratios ($\xi[\theta_n]$) is then computed over all test samples from the same direction θ_n to obtain the final estimated wideband beampattern. Similarly, by removing the frequency-based summation from both the numerator and denominator in (5), the narrowband beampattern is computed.

SDR: The signal-to-distortion ratio (SDR) [25], averaged over the test set, is used to measure the distortion in the estimated signals compared to the target signals.

5. EXPERIMENTAL RESULTS

5.1. Performance Analysis

To the best of our knowledge, three methods can perform spatial filtering exactly based on a predefined beampattern: the classic DMA [1], the NDF [15], and the parametric spatial filter [26]. For a fair comparison, NDF was trained for a dual-microphone array using the same training strategy and dataset as the NDBF.

As shown in Table 1, the proposed NDBF achieves the highest SDR. The performance of the DMA and the parametric spatial filter is significantly lower than that of the two neural network-based

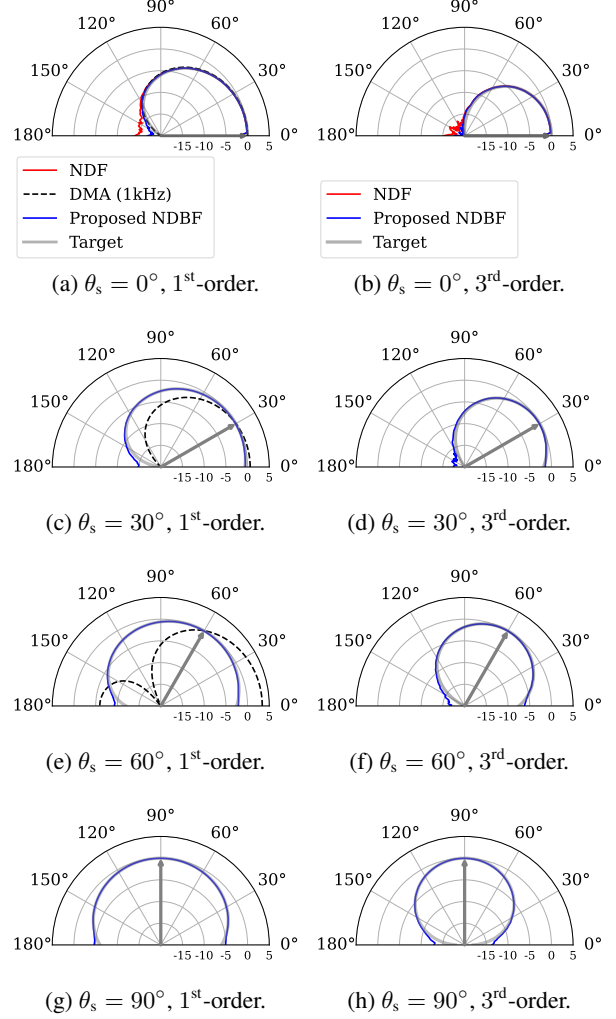


Fig. 4. Steerability study of the NDBF for various look directions θ_s and compared with the baselines.

methods. The DMA is subject to the well-known white-noise amplification problem at low frequencies [27] and spatial aliasing above 5.7 kHz due to an inter-microphone spacing of 3 cm, which substantially reduces its SDR. The parametric spatial filter is also impacted by spatial aliasing. To further compare NDF and the proposed NDBF, the estimated beampatterns produced by both methods were analyzed, as shown in Figs. 4(a) and (b) (wideband beampattern) and Figs. 5(a) and (b) (narrowband beampattern). Figs. 4(a) and (b) demonstrate that both methods effectively approximate the mainlobe of the target beampattern, while NDBF provides stronger suppression near the null position. This finding explains the SDR improvement of NDBF over NDF in Table 1, and also aligns with Figs. 5(a) and (b), which show that NDBF produces a cleaner suppression region in the narrowband beampattern. Additionally, both NDBF and NDF achieve a frequency-invariant beampattern without spatial aliasing for the broadband speech signals under test. The capability of NDF to achieve frequency-invariant beampattern has been previously analyzed in [18, 28], and the present study confirms that the proposed NDBF also exhibits this property.

Since the DMA does not guarantee a frequency-invariant beampattern, we analyze its beampattern at 1 kHz for comparison to investigate the steerability of the NDBF. We present estimated wide-

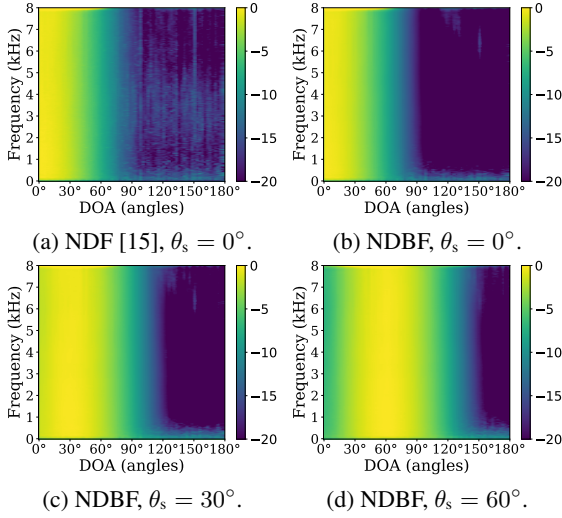


Fig. 5. Estimated 3rd-order narrowband beampatterns. (a) Baseline NDF [15] at $\theta_s = 0^\circ$. (b)-(d) Proposed NDBF at steering angles $\theta_s = 0^\circ, 30^\circ$, and 60° .

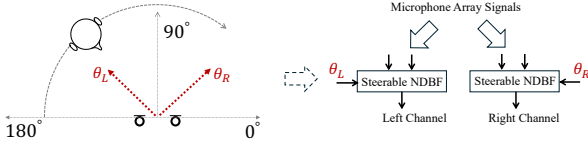


Fig. 6. Stereo recording using two omnidirectional microphones by the proposed steerable NDBF.

band beampatterns of NDBF for four different look directions (0° , 30° , 60° , and 90°) as illustrated in Fig. 4, where (a), (c), (e), and (g) corresponds to the 1st-order target beampattern and (b), (d), (f), and (h) corresponds to the 3rd-order target beampattern. The NDBF effectively approximates the target beampattern, and its beampattern remains consistent across different look directions, indicating that NDBF is steerable. In contrast, the DMA often achieves a response of 0 dB in the target direction but exceeds 0 dB in other directions, which is a typical non-steerable characteristic [3]. Figs. 5(c) and (d) show the estimated 3rd-order narrowband beampatterns of the NDBF for look directions at 30° and 60° . These results demonstrate that steerable high-order beampatterns maintain frequency-invariant properties similar to those observed in the endfire direction.

5.2. Application to Stereo Recording

We simulated a reverberant room with RT_{60} of 0.15 s and a size of 6 m $\times$ 4 m $\times$ 3.5 m. As shown in the left side of Fig. 6, a speech source moved steadily in a clockwise direction from 180° to 0° over approximately 12 seconds, with a fixed source-array distance of 1.5 m. The array consisted of two omnidirectional microphones spaced 3 cm apart. This simulated scene, generated using Dynamic Acoustic Scene Generator [29], is used to evaluate NDBF's stereo recording capability.

To the best of our knowledge, neither classical differential beamformers nor existing neural beamformers enable stereo recording with only two closely spaced omnidirectional microphones. Traditionally, stereo recording using the X-Y technique requires two first-order directional microphones, most commonly each with a cardioid

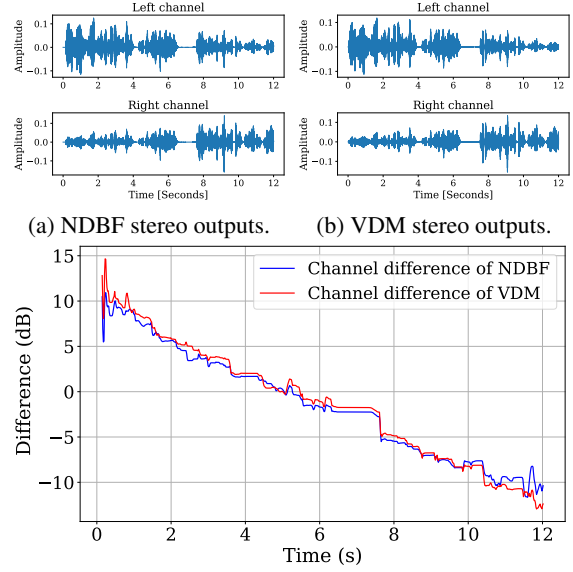


Fig. 7. Stereo output comparison between the proposed NDBF and desired virtual directional microphones (VDM).

beampattern, arranged at look directions of 45° and 135° [30]. In contrast, we trained a steerable NDBF model with a first-order cardioid beampattern as the training target. The right side of Fig. 6 illustrates a configuration in which two-microphone signals are processed by two parallel steerable NDBF models. One model is steered to $\theta_L = 135^\circ$, while the other is steered to $\theta_R = 45^\circ$. The resulting beamformed outputs are assigned to the left and right channels, respectively, thereby producing the stereo output of NDBF, shown in Fig. 7(a). For comparison, we simulate the X-Y technique using two virtual directional microphones (VDMs) located at the array center with the same look directions as the NDBF models. The stereo output of VDMs is presented in Fig. 7(b). This waveform comparison indicates that the stereo output of NDBF closely matches that of VDM. Figure 7(c) presents the segmental energy differences between the left and right channels for NDBF and VDM; the two curves are quite close, indicating highly similar inter-channel level differences. This qualitative example demonstrates that NDBF, even with a compact dual-microphone array, effectively captures the inter-channel level differences essential for stereo recording.

6. CONCLUSIONS

This paper presents a DNN-based differential beamformer, termed NDBF, which achieves steerable high-order DMA beampatterns using a dual-microphone array. Experimental results indicate that NDBF outperforms existing methods, particularly in approximating a target beampattern. This beampattern learning capability enables NDBF to overcome the limitations of traditional differential beamformers, which are typically first-order and non-steerable for such arrays. Furthermore, NDBF's steerability and beampattern-oriented spatial filtering enable stereo recording using only two closely spaced omnidirectional microphones.

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